

MH1200 Linear Algebra I.

Problem set #7.

SOLUTIONS

Core problems:

Problem 1:

Compute the inverses of the following matrices by using adjoint matrices.

$$\begin{pmatrix} 2 & 4 & 1 \\ 0 & 1 & 4 \\ 0 & 0 & 3 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & 0 \\ 3 & 0 & 3 \end{pmatrix}, \quad \begin{pmatrix} 1 & 3 & 5 \\ 2 & 1 & 4 \\ -1 & 6 & 2 \end{pmatrix}.$$

Solution

Solution: (i) The determinant of $\mathbf{A}$, obtained by expanding with regard to the 1st column, is

$$\det(\mathbf{A}) = +2 \begin{vmatrix} 1 & 4 \\ 0 & 3 \end{vmatrix} = 2 \cdot 3 = 6.$$

The cofactors of $\mathbf{A}$ are:

$$\begin{aligned} A_{11} &= + \begin{vmatrix} 1 & 4 \\ 0 & 3 \end{vmatrix} = 3, & A_{12} &= - \begin{vmatrix} 0 & 4 \\ 0 & 3 \end{vmatrix} = 0, & A_{13} &= + \begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix} = 0, \\ A_{21} &= - \begin{vmatrix} 4 & 1 \\ 0 & 3 \end{vmatrix} = -12, & A_{22} &= + \begin{vmatrix} 2 & 1 \\ 0 & 3 \end{vmatrix} = 6, & A_{23} &= - \begin{vmatrix} 2 & 4 \\ 0 & 0 \end{vmatrix} = 0, \\ A_{31} &= + \begin{vmatrix} 4 & 1 \\ 1 & 4 \end{vmatrix} = 15, & A_{32} &= - \begin{vmatrix} 2 & 1 \\ 0 & 4 \end{vmatrix} = -8, & A_{33} &= + \begin{vmatrix} 2 & 4 \\ 0 & 1 \end{vmatrix} = 2. \end{aligned}$$

From this we have

$$\mathbf{adj}(\mathbf{A}) = \begin{pmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{pmatrix}^T = \begin{pmatrix} 3 & 0 & 0 \\ -12 & 6 & 0 \\ 15 & -8 & 2 \end{pmatrix}^T = \begin{pmatrix} 3 & -12 & 15 \\ 0 & 6 & -8 \\ 0 & 0 & 2 \end{pmatrix}$$

and it follows from Theorem 2.5.30 in the Lecture Notes that

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \mathbf{adj}(\mathbf{A}) = \frac{1}{6} \begin{pmatrix} 3 & -12 & 15 \\ 0 & 6 & -8 \\ 0 & 0 & 2 \end{pmatrix}$$

(ii) The determinant of $\mathbf{B}$, obtained by expanding with regard to the 3rd column, is

$$\det(\mathbf{B}) = (-1)^{1+3} 3 \begin{vmatrix} 2 & -1 \\ 3 & 0 \end{vmatrix} + (-1)^{3+3} 3 \begin{vmatrix} 1 & 2 \\ 2 & -1 \end{vmatrix} = +3(3) + 3(-5) = -6.$$

After calculating the cofactors of $\mathbf{B}$ we find

$$\mathbf{B}^{-1} = \frac{1}{\det(\mathbf{B})} \mathbf{adj}(\mathbf{B}) = -\frac{1}{6} \begin{pmatrix} -3 & -6 & 3 \\ -6 & -6 & 6 \\ 3 & 6 & -5 \end{pmatrix}.$$

(iii) After calculating the cofactors of $\mathbf{C}$ we find

$$\mathbf{C}^{-1} = \frac{1}{\det(\mathbf{C})} \mathbf{adj}(\mathbf{C}) = \frac{1}{19} \begin{pmatrix} -22 & 24 & 7 \\ -8 & 7 & 6 \\ 13 & -9 & -5 \end{pmatrix}$$

□

Problem 2:

Let $\mathbf{A}$ be an invertible square matrix all of whose entries are integers. Show that all entries of $\mathbf{A}^{-1}$ are integers if and only if $\det(\mathbf{A}) = \pm 1$.

Comment: Please observe that because I wrote “if and only if” there are two logical directions in this statement for you to explain.

Solution

(i) Proof of the “if” part: Since the entries of $\mathbf{A}$ are integers it is clear that the cofactors A_{ij} are all integers as well: they are given by sums of products of integers. This implies that the entries of $\mathbf{adj}(\mathbf{A})$ are all integers. Then, if $\det(\mathbf{A}) = \pm 1$, it follows from the formula for the inverse

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \mathbf{adj}(\mathbf{A})$$

that the entries of $\mathbf{A}^{-1}$ are all integers.

(ii) Proof of the “only if” part: We need to show that when both the entries of $\mathbf{A}$ are all integers, and the entries of $\mathbf{A}^{-1}$ are all integers, then $\det(\mathbf{A}) = \pm 1$.

To see this, note that when the entries of $\mathbf{A}$ and $\mathbf{A}^{-1}$ are all integers then clearly the determinants $\det(\mathbf{A})$ and $\det(\mathbf{A}^{-1})$ are both integers as well. Then the relation

$$\det(\mathbf{A}) \det(\mathbf{A}^{-1}) = \det(\mathbf{A}\mathbf{A}^{-1}) = \det(\mathbf{I}) = 1$$

implies that $\det(\mathbf{A})$ and $\det(\mathbf{A}^{-1})$ must both be either +1 or -1 since these are the only integers whose product can give 1.

We have hereby shown that, when the entries of $\mathbf{A}$ are all integers, then the entries of $\mathbf{A}^{-1}$ can be all integers only if $\det(\mathbf{A}) = \pm 1$.

□

Problem 3:

Solve the following linear systems by applying Cramer's rule.

(i)

$$\begin{aligned}x + y - z &= 2 \\3x - y + z &= 5 \\3x + 2y + 4z &= 0\end{aligned}$$

Solution: The matrix form of the linear system is $\mathbf{A}\mathbf{u} = \mathbf{b}$ where

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & -1 \\ 3 & -1 & 1 \\ 3 & 2 & 4 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 2 \\ 5 \\ 0 \end{pmatrix}$$

Then Cramer's rule says that $x = \det(\mathbf{A}_1)/\det(\mathbf{A})$, $y = \det(\mathbf{A}_2)/\det(\mathbf{A})$ and $z = \det(\mathbf{A}_3)/\det(\mathbf{A})$ where $\mathbf{A}_i$ denotes the matrix obtained by replacing the i 'th column of $\mathbf{A}$ by $\mathbf{b}$. Calculating these determinants we find

$$\det(\mathbf{A}) = -24$$

and

$$\det(\mathbf{A}_1) = \begin{vmatrix} \mathbf{b} & 1 & -1 \\ 5 & -1 & 1 \\ 0 & 2 & 4 \end{vmatrix} = -42, \quad \det(\mathbf{A}_2) = \begin{vmatrix} 1 & \mathbf{b} & -1 \\ 3 & 5 & 1 \\ 3 & 0 & 4 \end{vmatrix} = 17, \quad \det(\mathbf{A}_3) = \begin{vmatrix} 1 & 1 & \mathbf{b} \\ 3 & -1 & 5 \\ 3 & 2 & 0 \end{vmatrix} = 23$$

Hence $x = -42/(-24) = 7/4$, $y = -17/24$, $z = -23/24$.

(ii)

$$\begin{aligned}x_1 - x_2 + x_3 + x_4 &= 4 \\x_1 + x_2 - x_3 - x_4 &= 0 \\-x_1 + x_2 + x_3 - x_4 &= -2 \\-x_1 + x_2 - x_3 + x_4 &= 4\end{aligned}$$

Solution

The linear system in this case is $\mathbf{A}\mathbf{x} = \mathbf{b}$ where

$$\mathbf{A} = \begin{pmatrix} 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ -1 & 1 & 1 & -1 \\ -1 & 1 & -1 & 1 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 4 \\ 0 \\ -2 \\ 4 \end{pmatrix}$$

The determinants we need to calculate in this case are

$$\begin{aligned}\det(\mathbf{A}) &= \begin{vmatrix} 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ -1 & 1 & 1 & -1 \\ -1 & 1 & -1 & 1 \end{vmatrix} = 8, & \det(\mathbf{A}_1) &= \begin{vmatrix} \mathbf{4} & -1 & 1 & 1 \\ \mathbf{0} & 1 & -1 & -1 \\ -\mathbf{2} & 1 & 1 & -1 \\ \mathbf{4} & 1 & -1 & 1 \end{vmatrix} = 16, \\ \det(\mathbf{A}_2) &= \begin{vmatrix} 1 & \mathbf{4} & 1 & 1 \\ 1 & \mathbf{0} & -1 & -1 \\ -1 & -\mathbf{2} & 1 & -1 \\ -1 & \mathbf{4} & -1 & 1 \end{vmatrix} = 24, & \det(\mathbf{A}_3) &= \begin{vmatrix} 1 & -1 & \mathbf{4} & 1 \\ 1 & 1 & \mathbf{0} & -1 \\ -1 & 1 & -\mathbf{2} & -1 \\ -1 & 1 & \mathbf{4} & 1 \end{vmatrix} = 8, \\ \det(\mathbf{A}_4) &= \begin{vmatrix} 1 & -1 & 1 & \mathbf{4} \\ 1 & 1 & -1 & \mathbf{0} \\ -1 & 1 & 1 & -\mathbf{2} \\ -1 & 1 & -1 & \mathbf{4} \end{vmatrix} = 32.\end{aligned}$$

Cramer's rule then gives

$$x_1 = \frac{16}{8} = 2, \quad x_2 = \frac{24}{8} = 3, \quad x_3 = \frac{8}{8} = 1, \quad x_4 = \frac{32}{8} = 4.$$

□

Problem 4:

Let

$$\mathbf{A} = \begin{pmatrix} 8 & a & 0 \\ 0 & 8 & a \\ a & 0 & 8 \end{pmatrix}$$

where a is a real number.

- (i) Compute $\det(\mathbf{A})$. For what values of a is $\mathbf{A}$ invertible?
- (ii) Under the condition that $\mathbf{A}$ is invertible, solve the matrix equation

$$\mathbf{A} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} b \\ c \\ d \end{pmatrix}$$

where b, c, d are real numbers. (Hint: The quickest way to do this now we have calculated the determinant of $\mathbf{A}$ is using Cramer's Rule.)

Solution

- (i) Cofactor expansion along the 1st row gives

$$\begin{vmatrix} 8 & a & 0 \\ 0 & 8 & a \\ a & 0 & 8 \end{vmatrix} = 8 \begin{vmatrix} 8 & a \\ 0 & 8 \end{vmatrix} - a \begin{vmatrix} 0 & a \\ a & 8 \end{vmatrix} = 8^3 + a^3.$$

By a theorem from the lectures, A is invertible if and only if $\det(A) \neq 0$. So A is invertible when $8^3 + a^3 \neq 0$, i.e. $a \neq -8$.

(ii) The easiest way to find the solution is to use Cramer's rule:

$$x_1 = \frac{\det(A_1)}{\det(A)} , \quad x_2 = \frac{\det(A_2)}{\det(A)} , \quad x_3 = \frac{\det(A_3)}{\det(A)}$$

where

$$\begin{aligned} \det(A_1) &= \begin{vmatrix} b & a & 0 \\ c & 8 & a \\ d & 0 & 8 \end{vmatrix} = b \begin{vmatrix} 8 & a \\ 0 & 8 \end{vmatrix} - c \begin{vmatrix} 8 & 0 \\ 0 & 8 \end{vmatrix} + d \begin{vmatrix} a & 0 \\ 8 & a \end{vmatrix} \\ &= 8^2b - 8ac + a^2d \\ \det(A_2) &= \begin{vmatrix} 8 & b & 0 \\ 0 & c & a \\ a & d & 8 \end{vmatrix} = -b \begin{vmatrix} 0 & a \\ a & 8 \end{vmatrix} + c \begin{vmatrix} 8 & 0 \\ a & 8 \end{vmatrix} + d \begin{vmatrix} 8 & 0 \\ 0 & a \end{vmatrix} \\ &= a^2b + 8^2c - 8ad \\ \det(A_3) &= \begin{vmatrix} 8 & a & b \\ 0 & 8 & c \\ a & 0 & d \end{vmatrix} = b \begin{vmatrix} 0 & 8 \\ a & 0 \end{vmatrix} - c \begin{vmatrix} 8 & a \\ a & 0 \end{vmatrix} + d \begin{vmatrix} 8 & a \\ 0 & 8 \end{vmatrix} \\ &= -8ab + a^2c + 8^2d \end{aligned}$$

This gives

$$x_1 = \frac{8^2b - 8ac + a^2d}{8^3 + a^3} , \quad x_2 = \frac{a^2b + 8^2c - 8ad}{8^3 + a^3} , \quad x_3 = \frac{-8ab + a^2c + 8^2d}{8^3 + a^3} .$$

□

Remark: This is not the only way to solve the problem. Another way is to compute A^{-1} , e.g. from $A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$, and then find the solution from $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = A^{-1} \begin{pmatrix} b \\ c \\ d \end{pmatrix}$. Another way is to consider the augmented matrix $\left(A \mid \begin{smallmatrix} b \\ c \\ d \end{smallmatrix} \right)$, reduce it to row-echelon form, and then solve by back-substitution. But both of these ways are longer/harder than using Cramer's rule as above.

Problem 5:

Let

$$\mathbf{A} = \begin{pmatrix} a & 0 & 0 & b \\ 0 & a & b & 0 \\ 0 & b & a & 0 \\ b & 0 & 0 & a \end{pmatrix},$$

where a and b are real numbers.

- (i) Evaluate $\det(\mathbf{A})$. For what values of a and b is $\mathbf{A}$ invertible?
- (ii) Compute the adjoint matrix of $\mathbf{A}$ for all values of a and b , and find $\mathbf{A}^{-1}$ for all values of a and b for which $\mathbf{A}$ is invertible.
- (iii) Consider the matrix equation

$$\mathbf{A}\mathbf{X}\mathbf{A} = \mathbf{I},$$

where $\mathbf{I}$ is an identity matrix. Determine the values of a and b for which a solution $\mathbf{X}$ exists (justify your answer), and solve the matrix equation in these cases.

Solution to Part (i).

$$a[a(a^2 - b^2)] - b[b(a^2 - b^2)]$$

Cofactor expansion along the 1st row gives

$$a^2[a^2 - b^2] - b^2[a^2 - b^2]$$

$$\det(A) = a \begin{vmatrix} a & b & 0 \\ b & a & 0 \\ 0 & 0 & a \end{vmatrix} - b \begin{vmatrix} 0 & a & b \\ 0 & b & a \\ b & 0 & 0 \end{vmatrix} = a^2(a^2 - b^2) - b^2(a^2 - b^2) = (a^2 - b^2)^2.$$

Next we must answer: For what values of a and b is A invertible?

By a theorem from the lectures, A invertible $\Leftrightarrow \det(A) \neq 0$. It follows from the result of (i) that A is invertible for all values of a and b that satisfy $a^2 \neq b^2$, i.e. $a \neq \pm b$.

Solution to Part (ii).

The (i, j) -entry of the adjoint matrix is the (j, i) -cofactor $A_{ji} = (-1)^{j+i} \det(M_{ji})$. Note that A is symmetric, therefore the adjoint matrix is also symmetric: $A_{ij} = A_{ji}$. So we only need to calculate A_{ij} for $i \leq j$. The calculations give:

$$A_{11} = A_{22} = A_{33} = A_{44} = a(a^2 - b^2)$$

$$A_{12} = A_{13} = A_{24} = A_{34} = 0$$

$$A_{14} = A_{23} = -b(a^2 - b^2).$$

$$A_{11} = (-1)^{1+1} \begin{vmatrix} a & b & 0 \\ b & a & 0 \\ 0 & 0 & a \end{vmatrix}$$

$$= a \begin{vmatrix} a & 0 \\ 0 & a \end{vmatrix} - b \begin{vmatrix} b & 0 \\ 0 & a \end{vmatrix}$$

$$= a(a^2) - b(ab)$$

$$= a[a^2 - b^2]$$

Hence

$$\text{adj}(A) = (a^2 - b^2) \begin{pmatrix} a & 0 & 0 & -b \\ 0 & a & -b & 0 \\ 0 & -b & a & 0 \\ -b & 0 & 0 & a \end{pmatrix}.$$

Then, according to a theorem in the lectures, when A is invertible its inverse is given by

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A) = \frac{1}{a^2 - b^2} \begin{pmatrix} a & 0 & 0 & -b \\ 0 & a & -b & 0 \\ 0 & -b & a & 0 \\ -b & 0 & 0 & a \end{pmatrix}.$$

Solution to Part (iii).

First note that AXA is only well-defined if X has the same number of rows and columns as A , so any solution X must be a 4×4 matrix. Next note that $I = AXA \Rightarrow 1 = \det(I) = \det(AXA) = \det(X) \det(A)^2$ where the product formula for determinants was used. This cannot hold if $\det(A) = 0$, i.e. $a^2 = b^2$, since then we would have $1 = 0$. So there is no solution X in this case. On the other hand, when $a^2 \neq b^2$ then A is invertible as noted earlier, and then there is a solution for X : Multiplying $AXA = I$ by A^{-1} from the left and right gives $X = A^{-1}IA^{-1} = (A^{-1})^2$.

Calculating this using the result for A^{-1} in part (iv), we find

$$X = (A^{-1})^2 = \frac{1}{(a^2 - b^2)^2} \begin{pmatrix} a^2 + b^2 & 0 & 0 & -2ab \\ 0 & a^2 + b^2 & -2ab & 0 \\ 0 & -2ab & a^2 + b^2 & 0 \\ -2ab & 0 & 0 & a^2 + b^2 \end{pmatrix}.$$

$$\begin{pmatrix} a & 0 & 0 & -b \\ 0 & a & -b & 0 \\ 0 & -b & a & 0 \\ -b & 0 & 0 & a \end{pmatrix} \cdot \begin{pmatrix} a & 0 & 0 & -b \\ 0 & a & -b & 0 \\ 0 & -b & a & 0 \\ -b & 0 & 0 & a \end{pmatrix}$$

□

$$= \begin{pmatrix} a^2 + b^2 & 0 & 0 & -2ab \\ 0 & a^2 + b^2 & -2ab & 0 \\ 0 & -2ab & a^2 + b^2 & 0 \\ -2ab & 0 & 0 & a^2 + b^2 \end{pmatrix}$$

Problem 6:

(A Quiz problem from 2019.)

Consider the following matrix $\mathbf{A}$, where a is a constant real number.

$$A = \begin{bmatrix} 1 & 1+a & 0 \\ 0 & 2 & 0 \\ a & a & 1 \end{bmatrix}.$$

- (a) For what values of the constant a is A invertible?
- (b) Calculate $\text{adj}(A)$, the adjoint of A .
- (c) Use part (b) to solve the equation

$$A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}.$$

Solution to (a).

For this part we'll use the theorem that a matrix is invertible if and only if the determinant is non-zero. (Although note that Gauss-Jordan elimination to reduced row echelon form is also very easy.)

The determinant can be calculated by a cofactor expansion down the first column. We deduce:

$$|A| = 1 * \begin{vmatrix} 2 & 0 \\ a & 1 \end{vmatrix} - 0 * \begin{vmatrix} 1+a & 0 \\ a & 1 \end{vmatrix} + a * \begin{vmatrix} 1+a & 0 \\ 2 & 0 \end{vmatrix} = 2.$$

This is always non-zero so we deduce that the matrix is invertible no matter what values a takes.

Solution to (b).

To calculate the adjoint we just apply the formula:

$$\begin{aligned}
 \text{adj}(A) &= \begin{bmatrix} \begin{vmatrix} 2 & 0 \\ a & 1 \end{vmatrix} & -\begin{vmatrix} 0 & 0 \\ a & 1 \end{vmatrix} & \begin{vmatrix} 0 & 2 \\ a & a \end{vmatrix} \\ -\begin{vmatrix} 1+a & 0 \\ a & 1 \end{vmatrix} & \begin{vmatrix} 1 & 0 \\ a & 1 \end{vmatrix} & -\begin{vmatrix} 1 & 1+a \\ a & a \end{vmatrix} \\ \begin{vmatrix} 1+a & 0 \\ 2 & 0 \end{vmatrix} & -\begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix} & \begin{vmatrix} 1 & 1+a \\ 0 & 2 \end{vmatrix} \end{bmatrix}^T \\
 &= \begin{bmatrix} 2 & 0 & -2a \\ -(1+a) & 1 & a^2 \\ 0 & 0 & 2 \end{bmatrix}^T \\
 &= \begin{bmatrix} 2 & -(1+a) & 0 \\ 0 & 1 & 0 \\ -2a & a^2 & 2 \end{bmatrix}
 \end{aligned}$$

Solution to (c).

The equation we need to solve is:

$$A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}.$$

Multiplying both sides by A^{-1} the solution can be expressed:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = A^{-1} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}.$$

The inverse is related to the adjoint by: $A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$. Using this we express the solution:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2 & -(1+a) & 0 \\ 0 & 1 & 0 \\ -2a & a^2 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}.$$

Thus:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2 + (1+a) \\ -1 \\ -2a - a^2 + 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2}(3+a) \\ -\frac{1}{2} \\ -\frac{a^2}{2} - a + 1 \end{bmatrix}.$$

□

Problem 7:

Let $\mathbf{A}$ be a n -by- n matrix. Let $\mathbf{B}$ be the matrix where the (i, j) entry $\mathbf{B}(i, j)$ is determined by the corresponding (i, j) entry of $\mathbf{A}$ via the formula $\mathbf{B}(i, j) = \frac{i}{j}\mathbf{A}(i, j)$. For example, in the 3-by-3 case, if

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} a_{11} & \frac{1}{2}a_{12} & \frac{1}{3}a_{13} \\ 2a_{21} & a_{22} & \frac{2}{3}a_{23} \\ 3a_{31} & \frac{3}{2}a_{32} & a_{33} \end{bmatrix}.$$

How does the determinant of $\mathbf{B}$ relate to that of $\mathbf{A}$?

Solution

Notice that we can obtain $\mathbf{B}$ from $\mathbf{A}$ by

1. Multiplying every row R_i by the corresponding row index i .
2. Multiplying every column C_j by $\frac{1}{j}$ which is one over the corresponding column index j .

Thus because we know that the determinant of a matrix $\mathbf{A}'$ we get by multiplying a row or column of a matrix $\mathbf{A}$ by a scalar c is $\det(\mathbf{A}') = c \det(\mathbf{A})$ we can deduce immediately that

$$\det(\mathbf{B}) = 3 * 2 * 1 * \frac{1}{1} * \frac{1}{2} * \frac{1}{3} * \det(\mathbf{A}) = \det(\mathbf{A}).$$

□

Problem 8:

Let $\mathbf{B}$ be an arbitrary $m \times n$ matrix.

- (i) Show that the matrix $\mathbf{B}^T \mathbf{B}$ is a symmetric matrix.
- (ii) Is it possible for every symmetric matrix $\mathbf{A}$ to be written as $\mathbf{B}^T \mathbf{B}$ for some matrix $\mathbf{B}$ of the same shape? Justify your answer.

Solution to (i).

To show a matrix is symmetric we need to show it is equal to its transpose. Observe:

$$(\mathbf{B}^T \mathbf{B})^T = \mathbf{B}^T (\mathbf{B}^T)^T = \mathbf{B}^T \mathbf{B}.$$

Solution to (ii).

No it's not always possible. If we take the determinant of the two sides of the equation $\mathbf{A} = \mathbf{B}^T \mathbf{B}$ we deduce:

$$\det(\mathbf{A}) = \det(\mathbf{B}^T \mathbf{B}) = \det(\mathbf{B}) \det(\mathbf{B}^T) = (\det(\mathbf{B}))^2.$$

So if the determinant of $\mathbf{A}$ is a negative number, it isn't possible. For example the following matrix is symmetric and has a negative determinant:

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}.$$

□

Problem 9: (Proof of Cramer's Rule.)

In this exercise we will discover an easy proof of Cramer's rule. Let $\mathbf{A}$ be an invertible $n \times n$ matrix, and let $\mathbf{B}$ be a $n \times 1$ matrix.

Because $\mathbf{A}$ is invertible the matrix equation $\mathbf{AX} = \mathbf{B}$ has a unique solution $\mathbf{X}$. Denote the components of the unique solution $\mathbf{X}$ via

$$\mathbf{X} = \begin{bmatrix} x_1 & x_2 & \dots & x_n \end{bmatrix}^T.$$

- (a) For $1 \leq i \leq n$ let $\mathbf{C}_i$ denote the $n \times 1$ matrix obtained from the i -th column of $\mathbf{A}$. Show that

$$\mathbf{B} = x_1 \mathbf{C}_1 + x_2 \mathbf{C}_2 + \dots + x_n \mathbf{C}_n. \quad (\star)$$

(There are various ways to see this. One interesting way is using the “splitting a matrix product up into blocks” trick.)

- (b) Let $\mathbf{A}_j$ denote the matrix obtained from $\mathbf{A}$ by replacing the j -th column with the matrix $\mathbf{B}$. Use fundamental properties of determinants to prove that

$$\det(\mathbf{A}_j) = x_j \det(\mathbf{A}).$$

- (c) Deduce Cramer's rule.

Solution to part (a)

To use the splitting into blocks trick: We organize $\mathbf{A}$ into a $1 \times n$ matrix of columns and the right hand side $\mathbf{B}$ into a 1×1 matrix whose entry is the column vector $\mathbf{B}$:

$$\mathbf{A} = \begin{bmatrix} \uparrow & & \uparrow \\ \mathbf{C}_1 & \dots & \mathbf{C}_n \\ \downarrow & & \downarrow \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = [\mathbf{B}].$$

The equality is exactly what you get when you compare the single entry of the two 1×1 sides of this equation.

Another method of solution is to compare the corresponding entries on the two sides of $(\star)$ directly.

Solution to part (b)

If we replace column j of $\mathbf{A}$ with $\mathbf{B}$ then compute the determinant we get

$$\det(\mathbf{A}_j) = \begin{vmatrix} \uparrow & & \uparrow & & \uparrow \\ \mathbf{C}_1 & \dots & \sum_{i=1}^n x_i \mathbf{C}_i & \dots & \mathbf{C}_n \\ \downarrow & & \downarrow & & \downarrow \end{vmatrix}.$$

Now we can use the linearity of determinant with respect to specific columns to expand this determinant into the following sum:

$$\det(\mathbf{A}_j) = \sum_{i=1}^n x_i \begin{vmatrix} \uparrow & & \uparrow & & \uparrow \\ \mathbf{C}_1 & \dots & \mathbf{C}_i & \dots & \mathbf{C}_n \\ \downarrow & & \downarrow & & \downarrow \end{vmatrix}.$$

In every one of these terms except the j -th term, the determinant has a repeated column and is zero. The remaining non-zero term is exactly:

$$x_j \det(\mathbf{A}).$$

That completes Part (b). Part (c) is a trivial rearrangement of the conclusion of Part (b).

□

Problem 10:

(From final exam 2019.)

- (a) Let $\mathbf{A}$ be an $n \times n$ matrix with the property that $\mathbf{A}^m = \mathbf{0}_{n \times n}$ where m, n are certain integers greater than zero. Calculate the determinant of $\mathbf{A}$. Justify your answer.
- (b) Let $\mathbf{B}$ be an arbitrary $m \times m$ matrix for some integer m greater than zero. Carefully derive a formula for $\det(\text{adj}(\mathbf{B}))$ in terms of $\det(\mathbf{B})$.

Solution to (a).

If we take the determinant of both sides of the equation

$$\mathbf{A}^m = \mathbf{0}_{n \times n}$$

we deduce

$$\det(\mathbf{A}^m) = \det(\mathbf{0}_{n \times n})$$

By multiplicativity of the determinant the LHS becomes $\det(\mathbf{A})^m$. And because the determinant of the zero matrix is zero we the RHS is zero.

Thus:

$$(\det(\mathbf{A}))^m = 0.$$

Thus:

$$\det(\mathbf{A}) = 0.$$

Solution to (b).

The starting point is the formula we proved in lectures:

$$\mathbf{B} \text{adj}(\mathbf{B}) = \det(\mathbf{B})\mathbf{I}.$$

If we take the determinant of both sides and use (i) the multiplicativity of the determinant (ii) the calculation that the determinant of $c\mathbf{I}_m$ is c^m then we deduce the formula

$$\det(\mathbf{B})\det(\text{adj}(\mathbf{B})) = \det(\mathbf{B})^m.$$

This implies:

$$\det(\mathbf{B}) (\det(\mathbf{B})^{m-1} - \det(\text{adj}(\mathbf{B}))) = 0.$$

There are two cases.

Case 1: $\det(\mathbf{B}) \neq 0$. In this case we can cancel $\det(\mathbf{B})$ and we deduce

$$\det(\operatorname{adj}(\mathbf{B})) = \det(\mathbf{B})^{m-1}.$$

Case 2: $\det(\mathbf{B}) = 0$. This case is more delicate. In this case the adjoint formula becomes

$$\mathbf{B} \operatorname{adj}(\mathbf{B}) = \mathbf{0}.$$

This implies $\operatorname{adj}(\mathbf{B})$ is not invertible. (If it was invertible we would deduce from this formula that $\mathbf{B} = \mathbf{0}$ which would imply $\operatorname{adj}(\mathbf{B}) = \mathbf{0}$ which would contradict invertibility.) Thus $\det(\operatorname{adj}(\mathbf{B})) = 0$.

In summary: The following formula is correct in both cases:

$$\det(\operatorname{adj}(\mathbf{B})) = \det(\mathbf{B})^{m-1}.$$

□

Problem 11.

(From final exam 2022.)

Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ -matrices for some $n \in \mathbb{N}$.

- (a) Prove that $(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$.
- (b) Let $\mathbf{A}$ be an invertible and symmetric matrix. Prove that $\mathbf{A}^{-1}$ is also a symmetric matrix.
- (c) Is the adjoint matrix of a symmetric matrix also symmetric? Briefly justify.

Solution to (a)

We just check corresponding entries of the two sides of the equation.

Consider some arbitrary $1 \leq i \leq n$ and some $1 \leq j \leq n$.

The (i, j) -entry of the left-hand-side $(\mathbf{AB})^T$ is by definition the (j, i) -entry of $\mathbf{AB}$. That equals

$$\sum_{k=1}^n a_{jk} b_{ki}.$$

On the other hand: The (i, j) -entry of the right-hand-side is

$$\sum_{k=1}^n (\mathbf{B}^T)_{ik} (\mathbf{A}^T)_{kj} = \sum_{k=1}^n (b)_{ki} (a)_{jk}.$$

These two expressions are equal so the identity is established.

Solution to (b)

The assumption that $\mathbf{A}$ is symmetric means that $\mathbf{A}^T = \mathbf{A}$.

We need to show that it follows that $(\mathbf{A}^{-1})^T = \mathbf{A}^{-1}$.

Take the equation: $\mathbf{AA}^{-1} = \mathbf{I}$. If we take the transpose of both sides, using Part (a), we deduce

$$(\mathbf{A}^{-1})^T \mathbf{A}^T = \mathbf{I}^T = \mathbf{I}.$$

Because $\mathbf{A}$ is symmetric this equation becomes:

$$(\mathbf{A}^{-1})^T \mathbf{A} = \mathbf{I}.$$

Multiplying both sides on the right by $\mathbf{A}^{-1}$ we deduce

$$(\mathbf{A}^{-1})^T = \mathbf{A}^{-1}.$$

Solution to (c)

Yes. The adjoint matrix $\text{adj}(\mathbf{A})$ of a symmetric matrix $\mathbf{A}$ is symmetric. Let's check the (i, j) -entry of the transpose of $\text{adj}(\mathbf{A})$. By definition it equals:

$$(\text{adj}(\mathbf{A}))_{ji} = (-1)^{i+j} \det(M_{ji})$$

where M_{ji} is the result of cancelling row j and column i from matrix $\mathbf{A}$.

Observe that because $\mathbf{A}$ is symmetric, $M_{ji} = (M_{ij})^T$.

Thus $((\text{adj}(\mathbf{A}))^T)_{ij}$ equals

$$(-1)^{i+j} \det((M_{ij})^T) = (-1)^{i+j} \det(M_{ij}) = ((\text{adj}(\mathbf{A})))_{ij}.$$

Thus the adjoint of a symmetric matrix is symmetric.

□

Problem 12:

Let $\mathbf{A}$ and $\mathbf{B}$ be two $n \times n$ matrices with determinant 1 and let $\mathbf{X} = \mathbf{A}^{-1}\mathbf{B}$.

For each k such that $1 \leq k < n$ let $\mathbf{X}_k$ denote the $(n-k) \times (n-k)$ matrix that results from forgetting the first k rows and the last k columns of $\mathbf{X}$. (In words, $\mathbf{X}_k$ is the lower-left square $(n-k) \times (n-k)$ corner block appearing inside the matrix $\mathbf{X}$. It will probably be helpful to draw this on a diagram of the matrix.)

Let $\mathbf{Y}_k$ be the $n \times n$ matrix obtained by making the first k columns of $\mathbf{Y}_k$ the first k columns of $\mathbf{A}$, and making the remaining $(n-k)$ columns of $\mathbf{Y}$ the first $(n-k)$ columns of $\mathbf{B}$.

Prove that for all $1 \leq k < n$:

$$\det(\mathbf{X}_k) = \det(\mathbf{Y}_k)$$

Solution

For the discussion, let the columns of $\mathbf{A}$ be denoted A_1 up to A_n and let the columns of $\mathbf{B}$ denoted B_1 up to B_n .

We wish to calculate the determinant of the following matrix, where $1 \leq k < n$:

$$\mathbf{Y}_k = \begin{bmatrix} \uparrow & & \uparrow & \uparrow & & \uparrow \\ A_1 & \cdots & A_k & B_1 & \cdots & B_{n-k} \\ \downarrow & & \downarrow & \downarrow & & \downarrow \end{bmatrix}.$$

A trick to understanding this is to multiply this on the left by $\mathbf{A}^{-1}$. (Because $\mathbf{A}$ has non-zero determinant the inverse exists.)

By computing the entries explicitly it is possible to see directly that the result of this multiplication is:

$$\mathbf{A}^{-1}\mathbf{Y}_k = \begin{bmatrix} \mathbf{I}_k & \mathbf{M}_{k \times (n-k)} \\ \mathbf{0}_{(n-k) \times k} & \mathbf{X}_k \end{bmatrix}$$

where $\mathbf{M}_{k \times (n-k)}$ is some $k \times (n-k)$ matrix we don't need to know exactly and $\mathbf{X}_k$ is the $(n-k) \times (n-k)$ corner block of $\mathbf{X}$ introduced in the problem statement.

Now we just calculate the determinant of the two sides of this expression. The left-hand-side is computed:

$$\det(\mathbf{A}^{-1}\mathbf{Y}_k) = \det(\mathbf{A})^{-1} \det(\mathbf{Y}_k) = \det(\mathbf{Y}_k)$$

using the given information that $\det(\mathbf{A}) = 1$.

On the other hand because the right-hand-side is block diagonal we know that its determinant equals the product of the determinants of the diagonal blocks:

$$\det(\mathbf{I}_k)\det(\mathbf{X}_k) = \det(\mathbf{X}_k).$$

Equating the two determinants we conclude:

$$\det(\mathbf{X}_k) = \det(\mathbf{Y}_k)$$

as required.

□